

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/16 Modified]

Two coherent waves from a double-slit meet on the screen to form an interference pattern. What is the phase difference of the two waves at the second-order minima?

- A 0 rad
- B $\frac{1}{2}\pi$ rad
- C π rad
- D $\frac{3}{2}\pi$ rad

2. [ACJC/H2/Prelims 2018/P1/18 Modified]

Two blue dots are projected on a screen. A student who is facing the screen takes a few steps backwards and stops walking when she could just resolve the two dots.

The student wishes to still resolve the two dots by standing further back from the screen.

What should she do?

- A Darken the room to increase the size of the pupil of the eye.
- B Change the colour of the two dots from blue to red.
- C Make the blue dots slightly brighter.
- D Move the two dots closer.

3. [AJC/H2/Prelims 2018/P1/15]

Two progressive waves of frequency 300 Hz are superimposed to produce a stationary wave. The distance between the first node and the sixth node is 7.5 m.

What is the speed of the progressive waves?

- A 375 m s⁻¹
- B 450 m s⁻¹
- C 750 m s⁻¹
- D 900 m s⁻¹

4. [AJC/H2/Prelims 2018/P1/16]

A camera lens with a maximum aperture of 25 mm forms an image of an object 9.0 m away.

If a monochromatic light of wavelength 500 nm is used, what is the minimum distance between two points on the object that are just resolved?

- A 0.09 mm
- B 0.18 mm
- C 5.2 mm
- D 10 mm

5. [CJC/H2/Prelims 2018/P1/15]

In a two-slit interference experiment, one slit transmits waves of twice the amplitude compared to the other slit. If the maximum intensity of the interference pattern is I_0 , what is the minimum intensity of the pattern?

- A Zero
- B $\frac{I_0}{2}$
- C $\frac{I_0}{4}$
- D $\frac{I_0}{9}$

6. [CJC/H2/Prelims 2018/P1/16]

A space station orbits at a height of 335 km above the surface of the Earth. It carries two panels separated by a distance of 25 m. The panels reflect light of wavelength 500 nm towards an observer on the Earth's surface.

The observer views the panels with a telescope that has an aperture diameter of 200 mm. Assume that the panels act as point sources of light for the observer.

Which of the following is correct?

	Will the two images seen by the observer be resolved?	Angular separation of two sources as measured from aperture / rad
A	Yes	2.5×10^{-6}
B	Yes	7.5×10^{-5}
C	No	2.5×10^{-6}
D	No	7.5×10^{-5}

7. [DHS/H2/Prelims 2018/P1/17]

S_1 and S_2 are loudspeakers facing each other and emitting continuous sound waves of frequency 1100 Hz. M is a small microphone which runs on a straight track between S_1 and S_2 at a speed of 30 m s⁻¹. The sound received by M will fluctuate with a frequency f .

If the velocity of sound is 330 m s⁻¹, what is the value of f ?

- A 100 Hz
- B 200 Hz
- C 400 Hz
- D 800 Hz

8. [DHS/H2/Prelims 2018/P1/18]

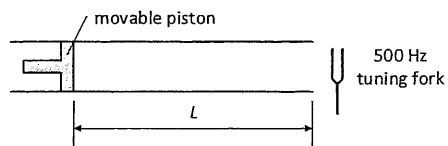
Light of wavelength 600 nm is incident on a pair of slits. Fringes with a spacing of 4.0 mm are formed on a screen.

What will be the fringe spacing when the wavelength of the light is changed to 400 nm and the separation of the slits is doubled?

- A 1.3 mm
- B 3.0 mm
- C 5.3 mm
- D 12 mm

9. [EJC/H2/Prelims 2018/P1/14]

A 500 Hz tuning fork is held at the open end of an air-filled glass tube, which is closed at the other end by a movable piston.



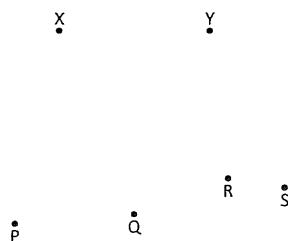
Resonance is achieved when $L = 82.5$ cm. Given that the speed of sound in air is 330 m s^{-1} , which of the following other values of L will not result in resonance?

- A 16.5 cm B 49.5 cm C 99.0 cm D 115.5 cm

10. [EJC/H2/Prelims 2018/P1/16 Modified]

In the diagram below (not drawn to scale), X and Y are identical point sources of waves that exhibit a constant phase difference of π at source. The waves have a wavelength of 1.0 cm.

Given $XP = 4.0$ cm; $YP = 5.5$ cm; $XQ = 4.0$ cm; $YQ = 4.0$ cm; $XR = 4.5$ cm; $YR = 3.0$ cm; $XS = 5.5$ cm; $YS = 3.5$ cm.

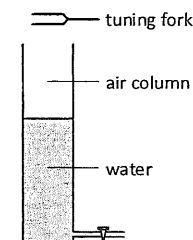


Which of the following statements is true?

- A P is a point of destructive interference.
 B $XQ - YQ = \frac{1}{2}m\lambda$, where m is an odd integer.
 C $XR - YR = n\lambda$, where n is an integer.
 D S is a point of destructive interference.

11. [HCI/H2/Prelims 2018/P1/14]

The diagram shows a setup in which a stationary wave is produced in an air column. A tuning fork, placed above the tube, vibrates and produces a sound wave. The length of the air column can be varied by altering the volume of water in the tube.



Initially, water is filled to the brim of this tube. The water is allowed to run out of it. Resonance occurs when the air column lengths are 18 cm and 30 cm. Which of the following lengths of air column will not result in resonance?

- A 6 cm B 24 cm C 42 cm D 54 cm

12. [HCI/H2/Prelims 2018/P1/15]

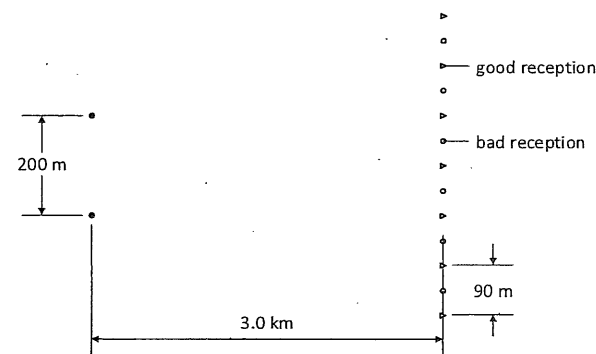
Light of wavelengths 400 nm and 600 nm are incident normally on a diffraction grating. It was observed that the 400 nm light in one order of the spectrum appears at the same angle as 600 nm light in the adjacent order.

Given that the angle is 30° , calculate the spacing between the slits in the grating.

- A $1.2 \mu\text{m}$ B $1.8 \mu\text{m}$ C $2.4 \mu\text{m}$ D $3.6 \mu\text{m}$

13. [IJC/H2/Prelims 2018/P1/17]

Two radio transmitters 200 m apart are transmitting on the same frequency as shown in the diagram below.



At a distance of 3.0 km away, it was found that there are series of good and poor areas of reception, where distance between consecutive good reception is 90 m apart.

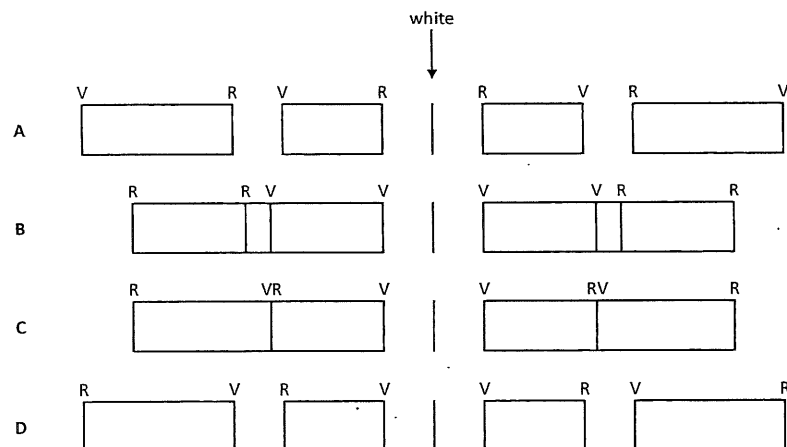
Which one of the following is the wavelength of the radio wave?

- A 0.17 m B 6.0 m C 670 m D 1350 m

14. [JJC/H2/Prelims 2018/P1/18]

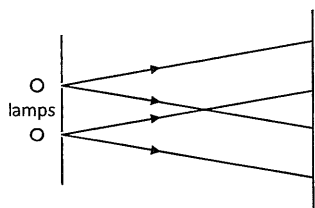
A narrow shaft of white light (from violet 350 nm (V) to red 700 nm (R)) falls with normal incidence on a diffraction grating and produces two orders of spectra on a distant screen.

Which of the following diagram shows the appearance of the spectra?



15. [JJC/H2/Prelims 2018/P1/21]

A student set up an interference effect experiment using two lamps emitting white light and a pair of slits as shown in the diagram below. However, no interference pattern was observed on a screen placed a distance away from the pair of slits.



Which of the following is the reason for the observation?

- A The lamps are not point sources.
- B The lamps emit light of different amplitudes.
- C The light from the lamps is not coherent.
- D The light from the lamps is white.

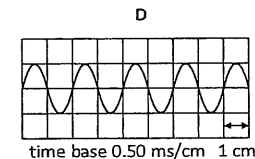
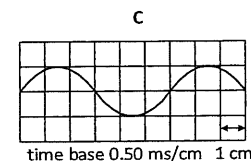
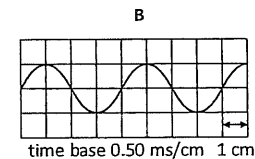
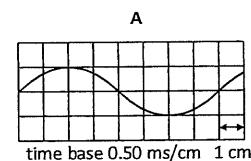
16. [JJC/H2/Prelims 2018/P1/22]

A stationary sound wave is set up between a loudspeaker and a wall.

A microphone is connected to a cathode-ray oscilloscope (c.r.o.) and is moved along a line directly between the loudspeaker and the wall. The amplitude of the trace on the c.r.o. rises to a maximum at a position X, falls to a minimum and then rises once again to a maximum at a position Y.

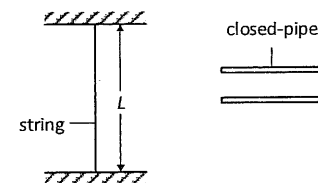
The distance between X and Y is 33 cm. The speed of sound in air is 330 m s^{-1} .

Which diagram represents the c.r.o. trace of the sound received at X?



17. [MI/H2/Prelims 2018/P1/20 Modified]

The figure shows a stretched string of length L placed near a closed-pipe. The fundamental mode of oscillation in the string occurs when the speed of waves in the string equals the speed of soundwaves in the air.



What is the length of the closed pipe that will resonate with the sound produced by the string?

- A $0.5L$
- B L
- C $2L$
- D $3L$

18. [MI/H2/Prelims 2018/P1/21]

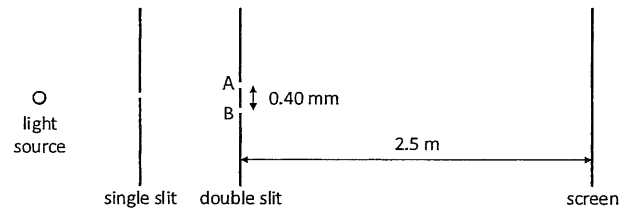
Red light of wavelength 700 nm passes through a Young's double-slit arrangement. Fringes of separation Z are observed in a plane 1.20 m away from the slits. The light source is then replaced with blue light of wavelength 420 nm.

At what distance from the slits would fringes of the same separation Z be observed?

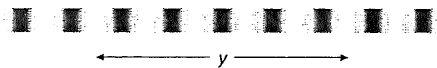
- A 0.60 m
- B 0.72 m
- C 1.67 m
- D 2.00 m

19. [MJC/H2/Prelims 2018/P1/11]

A two-source interference experiment is set up as shown.



The source emits light of wavelength 600 nm. The interference pattern on the screen is shown below.

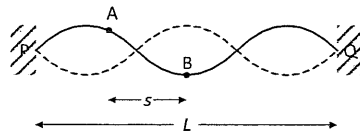


What is the distance y ?

- A 3.8×10^{-4} m B 1.9×10^{-3} m C 3.8×10^{-3} m D 1.9×10^{-2} m

20. [MJC/H2/Prelims 2018/P1/12]

A guitar string of length L is stretched between two fixed points P and Q and made to vibrate transversely as shown.



Two particles A and B on the string are separated by a distance s . The maximum kinetic energies of A and B are K_A and K_B respectively.

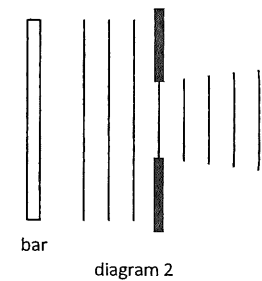
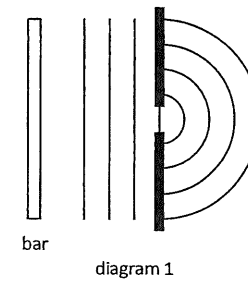
Which of the following gives the correct phase difference and maximum kinetic energies of the particles?

	phase difference	maximum kinetic energy
A	$\left(\frac{3s}{2L}\right) \times 360^\circ$	$K_A < K_B$
B	$\left(\frac{3s}{2L}\right) \times 360^\circ$	same
C	180°	$K_A < K_B$
D	180°	same

21. [MJC/H2/Prelims 2018/P1/13]

Diagram 1 shows a ripple tank experiment in which plane waves are diffracted through a narrow slit in a metal sheet. Diagram 2 shows the same tank with a slit of greater width.

In each case, the pattern of the incident wave fronts before the slit and emergent wave fronts after the slit are shown.



Which of the following changes would cause the waves in diagram 1 to produce an emergent pattern closer to that shown in diagram 2?

- A Increasing the frequency of vibration of the bar.
 B Increasing the speed of the waves by making the water in the tank deeper.
 C Reducing the amplitude of vibration of the bar.
 D Reducing the length of the vibrating bar.

22. [NJC/H2/Prelims 2018/P1/14]

A boy blows gently across the top of a piece of glass tubing the lower end of which is closed by his finger so that the tube gives its fundamental note of frequency f .

While blowing, he removes his finger from the lower end. The note he then hears will have a frequency of approximately

- A $f/4$ B $f/2$ C $2f$ D $4f$

23. [NJC/H2/Prelims 2018/P1/15]

Light of wavelength λ is incident on a pair of slits, forming fringes 3.0 mm apart on a screen.

What is the fringe spacing when light of wavelength 0.5λ is used and the slit separation is doubled?

- A 0.75 mm B 1.5 mm C 3.0 mm D 6.0 mm

24. [NJC/H2/Prelims 2018/P1/16 Modified]

A beam of monochromatic light is incident normally on a diffraction grating with N_1 lines per millimetre. The second order diffracted beam makes an angle of 45° with the grating.

The grating is then replaced by one with N_2 lines per millimetre. The third order diffracted beam is now observed at the same angle.

What is the ratio of N_1/N_2 ?

- A 0.47 B 0.67 C 1.1 D 1.5

25. [NYJC/H2/Prelims 2018/P1/16]

The two lowest frequencies at which an open narrow tube resonates are 256 Hz and 512 Hz.
If one of the ends is closed, what are the two lowest frequencies at which it would now resonate?

- A 128 Hz and 256 Hz
B 128 Hz and 384 Hz
C 384 Hz and 640 Hz
D 768 Hz and 1536 Hz

26. [NYJC/H2/Prelims 2018/P1/17]

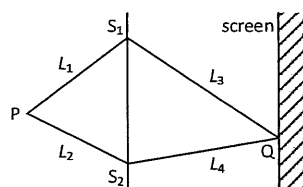
A beam of light that consists of all wavelengths between 480 nm and 600 nm is projected on a diffraction grating that contains 500 lines per millimeter.

What is the maximum number of complete continuous spectra that can be observed emerging from the grating?

- A 6 B 7 C 8 D 9

27. [PJC/H2/Prelims 2018/P1/14]

Two identical narrow slits S_1 and S_2 are illuminated by light of wavelength λ from a point source P. The light is then allowed to fall on a screen.



Given that m is a positive integer, what is the condition for destructive interference to occur at point Q?

- A $(L_3 - L_4) = m\lambda$
B $(L_3 - L_4) = \frac{1}{2}(2m+1)\lambda$
C $(L_1 + L_3) - (L_2 - L_4) = m\lambda$
D $(L_1 + L_3) - (L_2 + L_4) = \frac{1}{2}(2m+1)\lambda$

28. [PJC/H2/Prelims 2018/P1/15]

A wire is stretched between two fixed points 1.2 m apart. It is set into vibration while the mid-point of the wire is held in place.

Which are the possible wavelengths of the waves in the wire?

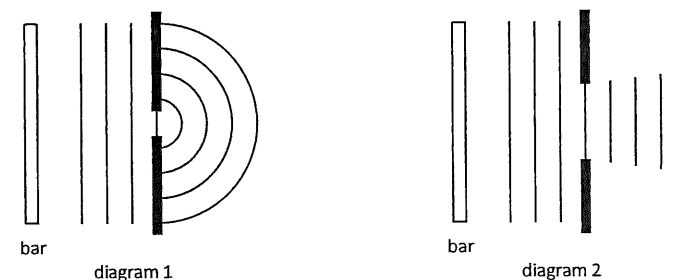
- A 0.20 m, 0.60 m, 2.4 m
B 0.40 m, 0.80 m, 1.2 m
C 0.24 m, 0.30 m, 0.60 m
D 0.60 m, 1.2 m, 2.4 m

29. [RVHS/H2/Prelims 2018/P1/15]

Diagram 1 shows a ripple tank experiment in which plane waves are diffracted through a narrow slit in a metal sheet.

Diagram 2 shows the same tank with a slit of greater width.

In each case, the pattern of the incident wave fronts before the slit and emergent wave fronts after the slit are shown.

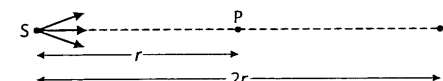


Which of the following changes would cause the waves in diagram 2 to produce an emergent pattern closer to that shown in diagram 1?

- A Decreasing the density of the liquid.
B Decreasing the frequency of vibration of the bar.
C Decreasing the speed of the waves by making the water in the tank shallower.
D Increasing the amplitude of vibration of the bar.

30. [SRJC/H2/Prelims 2018/P1/18]

A point source of sound is placed at point S.



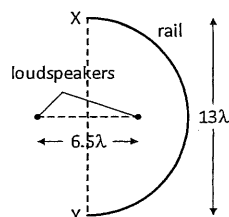
The air molecules at P, a distance r from S, oscillate with an amplitude of $8.0 \mu\text{m}$. Point Q is situated at a distance $2r$ from S.

What is the amplitude of oscillation of air molecules at point Q?

- A $1.4 \mu\text{m}$ B $2.0 \mu\text{m}$ C $2.8 \mu\text{m}$ D $4.0 \mu\text{m}$

31. [SRJC/H2/Prelims 2018/P1/19]

Two loudspeakers are emitting sound of wavelength λ in all directions. They are in phase with each other and are placed a distance 6.5λ apart in the middle of a semicircular rail of diameter 13λ , as shown.



Moveable microphones along the rail are used to detect the sound intensity along the rail. The midpoint of the line joining the 2 speakers coincides with the centre of line XY.

How many minima will the microphones detect?

- A 7 B 12 C 13 D 14

32. [TPJC/H2/Prelims 2018/P1/19]

When light of wavelength 570 nm is incident normally on a plane diffraction grating, the angle between second-order diffraction images is 69.6° .

What is the number of lines per millimetre of the grating?

- A 500 B 1000 C 2000 D 500000

33. [TPJC/H2/Prelims 2018/P1/20]

A standing wave is set up on a stretched string XY as shown.



At which point(s) will the oscillation be in phase with that at P?

- A 1 and 2 only B 1 only C 2 only D 3 only

34. [VJC/H2/Prelims 2018/P1/16]

A string of length 1.0 m is fixed at both ends and made to vibrate.

Which of the following wavelengths cannot be formed on the string?

- A 0.50 m B 0.67 m C 1.3 m D 2.0 m

35. [VJC/H2/Prelims 2018/P1/17]

A satellite orbits the Earth at a height of 120 km. It has a camera of aperture 2.5 cm.

By considering a wavelength of visible light of 5.5×10^{-7} m, what is the nearest distance between two objects on the surface of Earth that can be distinguished as separate objects?

- A 1.3 m B 2.6 m C 3.9 m D 5.2 m

36. [YJC/H2/Prelims 2018/P1/16]

A car passes you on the highway and you notice the tail lights of the car are 1.26 m apart. The car is 14.4 km away when the tail lights appear to merge into a single spot of light.

What is the angular resolution as the lights just appear to merge into a single spot of light?

- A 4.73×10^{-5} rad B 8.75×10^{-5} rad C 4.73×10^{-3} rad D 8.75×10^{-3} rad

37. [YJC/H2/Prelims 2018/P1/17]

In a Young's double-slit experiment, a coherent monochromatic light illuminates the two slits, producing an interference pattern on a screen some distance away.

Which modification increases the separation between the dark fringes on the screen?

- A Increasing the slit separation
B Decreasing distance between the screen and the slits
C Using monochromatic light of higher intensity
D Using monochromatic light of lower frequency

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P3/7 Modified]

- a. Explain what is meant by the phase difference of two waves of the same frequency. [1]
- b. An experiment to investigate two-source interference using light waves is set up as shown in Fig. 1.1. Three slits S_0 , S_1 and S_2 of equal width are used with a lamp that is attached with a red filter.



Fig. 1.1

- i. For observable interference, suggest why slit S_0 is needed when the light source used is a lamp. [2]
- ii. Draw, on Fig. 1.1, lines to represent the rays of light from S_0 to the point C on the screen which is equidistant to S_1 and S_2 . [2]
- iii. State the path difference of the light reaching point C. [1]
- iv. 1. The amplitude of the light at point C on the screen in Fig. 1.1 is A. State and explain the new amplitude at point C on the screen when a piece of polariser is placed in front of S_2 . [2]
2. Describe the changes to the interference pattern on the screen if the red filter in Fig. 1.1 is replaced with a blue filter. [2]
- c. i. Explain the formation of a stationary wave. [2]
- ii. A stationary wave is set up in a closed tube using a speaker connected to a signal generator and is as shown in Fig. 1.2.

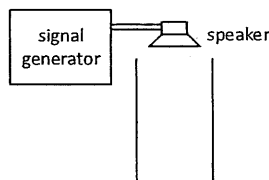


Fig. 1.2

The length of the tube is 90 cm long and the wavelength of the sound is 120 cm. A small microphone which senses pressure variation is placed in the tube to measure the frequency of the stationary wave formed.

The microphone is connected to a cathode-ray oscilloscope (c.r.o.). When the microphone picks up the largest pressure variation, the waveform obtained on the c.r.o. is shown in Fig. 1.3.

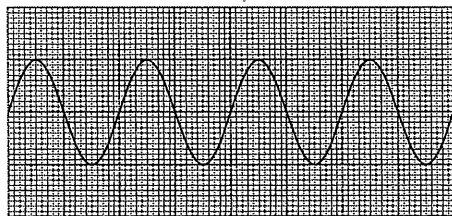


Fig. 1.3

1. State and explain the distance in the tube from the open end for the microphone to be placed so that the waveform in Fig. 1.3 is obtained. [3]
- Include in your answer a diagram of the stationary wave setup in the air column of the tube. Label your diagram. [3]
2. The wavelength of the sound from the speaker is increased to 330 cm. State and explain if a stationary wave can be formed. [2]
- d. A monochromatic laser is incident on a diffraction grating. By drawing a suitable well-labelled diagram, describe how you would determine the wavelength of the laser light. [4]

[Total: 21]

2. [AJC/H2/Prelims 2018/P3/5]

- a. State the principle of superposition. [2]
- b. Fig. 2.1 shows the arrangement for viewing a visible interference pattern on a screen.

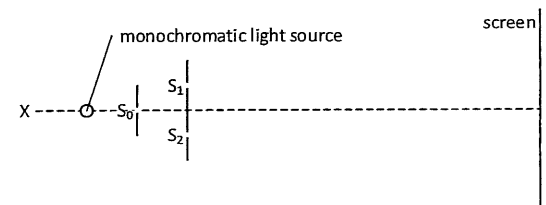


Fig. 2.1

In a darkened room, a double slit (S_1S_2) is placed in front of a narrow single slit S_0 situated in front of a monochromatic light source. Line XY is equidistant from S_1 and S_2 .

- i. The point Y on the screen is directly opposite to the centre of the double slit. State and explain the nature of the interference that occurs at Y. [2]
- ii. The distance between slits S_1 and S_2 is 0.60 mm. When the screen is placed 1.8 m from the slits, the distance between the two first-order bright fringes in the interference pattern formed on the screen is 4.0 mm. Calculate the wavelength of the light. [2]
- iii. Suggest changes to the appearance of the fringes when each of the following changes is made separately.
- Moving the light source and single slit along the axis towards point X, further away from S_1S_2 . [2]
 - Shifting the single slit a small distance downwards, closer to slit S_2 . [2]

[Total: 10]

3. [CJC/H2/Prelims 2018/P2/4]

- a. Describe one condition necessary for observable two-source interference fringes to be formed. [1]
- b. Two microwave transmitters producing waves of the same frequency are placed at P and Q which are at a distance of $2d$ apart. Points Y and Z are equidistant from O. The line YXZ is perpendicular to OX, as shown in Fig. 3.1.

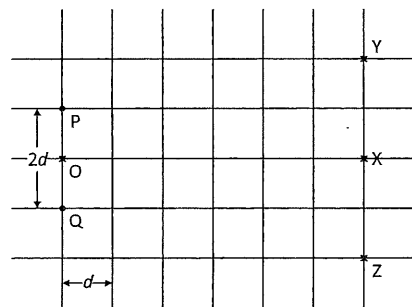


Fig. 3.1

- i. The waveforms of the microwaves from P and Q arriving at point Y vary with time as shown in Fig. 3.2.

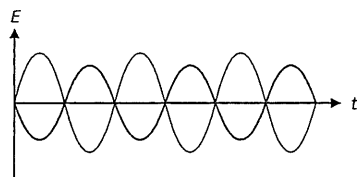


Fig. 3.2

1. State and explain if the waves arriving at Y are coherent. [2]
 2. Explain why a minimum intensity is detected at Y. [1]
- ii. Show that the path difference of the waves arriving at point Y from P and Q is $0.625d$. [1]
- iii. As a detector is moved along a straight line from X to Y, it encountered three intensity maxima, including the maximum at X. [3]
- Determine the frequency of the wave in terms of d .

[Total: 8]

4. [CJC/H2/Prelims 2018/P2/5]

In a simple experiment to find the wavelength of monochromatic red light emitted by a laser, a fine beam of red laser light is incident on a diffraction grating as shown in Fig. 4.1. The diffraction grating has 300 lines per millimeter and it is arranged such that its plane is normal to the incident light.

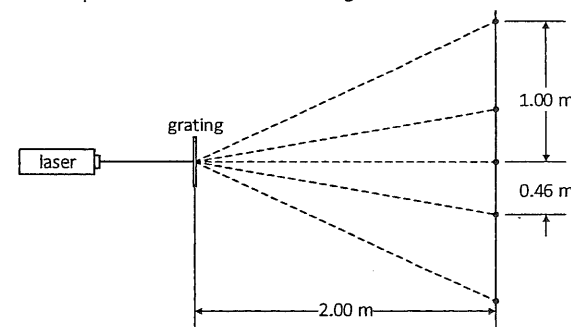


Fig. 4.1

Bright spots are observed at 0.46 m and 1.00 m from the central spot on a screen, which is 2.00 m from the grating.

- a. By considering the bright spot at 0.46 m from the central spot, calculate the wavelength of the laser light. [3]
- b. Suggest and explain an experimental advantage of obtaining the wavelength of the laser light by using the second-order diffracted light rather than the first-order diffracted light. [2]

[Total: 5]

5. [CJC/H2/Prelims 2018/P3/4]

A length of wire is held taut between two points X and Y as shown in Fig. 5.1. A signal generator which produces an alternating current of variable frequency is passed through the wire and a pair of magnets is placed on either side of the wire.

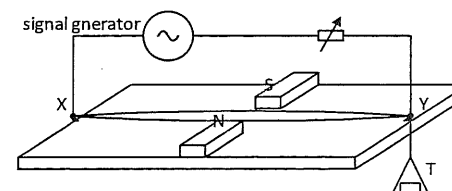


Fig. 5.1

The frequency of the alternating current is gradually increased from zero. A stationary wave is set up as shown in Fig. 5.1 when the frequency is 10 Hz.

- a. Explain how the stationary wave is formed on the wire when an alternating current is passed through it. [4]
- b. The distance between X and Y is 0.60 m.
Calculate the wavelength of the stationary wave formed. [1]
- c. Determine the speed of the wave. [2]
- d. Sketch the stationary wave formed in the space below when the frequency of the alternating current is adjusted to 30 Hz. [2]

[Total: 9]

6. [DHS/H2/Prelims 2018/P2/4 Modified]

- a. A narrow beam of light of wavelength 632 nm is incident normally on a diffraction grating as shown in Fig. 6.1.

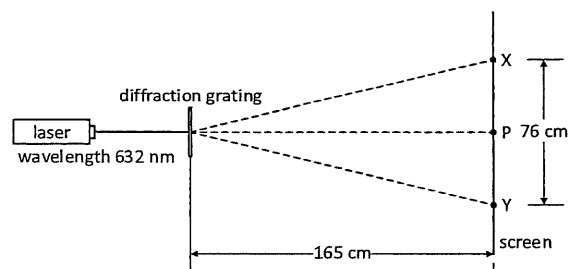


Fig. 6.1 (front view)

Spots of light are observed on a screen placed parallel to the grating. The distance between the grating and the screen is 165 cm.

The brightest spot is P. The spots formed closest to P and on each side of P are X and Y. X and Y are separated by a distance of 76 cm.

- Calculate the number of lines per metre on the grating. [3]
- Light of wavelengths 632 nm and 638 nm is now incident normally on the grating. Two lines are observed in the first order spectrum and two lines are observed in the second order spectrum, corresponding to the two wavelengths. [2]

State two differences between the first order spectrum and the second order spectrum.

- The grating in a. is now rotated about an axis parallel to the incident light, as shown in Fig. 6.2.

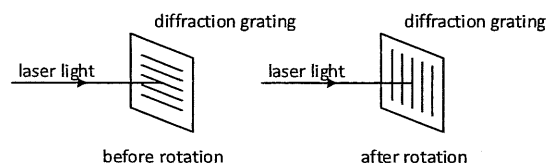


Fig. 6.2

State what effect, if any, this rotation will have on the positions of the spots P, X and Y. [2]

- In another experiment using the apparatus in a., it was noticed that the distances XP and PY, as shown in Fig. 6.1, are not equal.

Suggest a reason for this difference. [1]

[Total: 8]

7. [EJC/H2/Prelims 2018/P2/4 Modified]

- a. Noise-cancelling headphones use both Active Noise Control and passive sound-proofing to reduce undesired ambient sounds reaching the ears of the user. In Active Noise Control, a microphone detects ambient noise and a noise-cancellation speaker emits a corresponding “anti-noise” that undergoes destructive interference with the noise.

Suggest the features that the “anti-noise” should have in order to achieve the optimal noise-cancelling effect. [2]

- b. A student measured the waveforms of the noise and the anti-noise. The following observations were made:
- There is a 1.0 ms time-lag between the noise and the anti-noise.
 - The amplitudes of the noise and anti-noise were effectively similar.

A sample noise waveform is shown in Fig. 7.1.

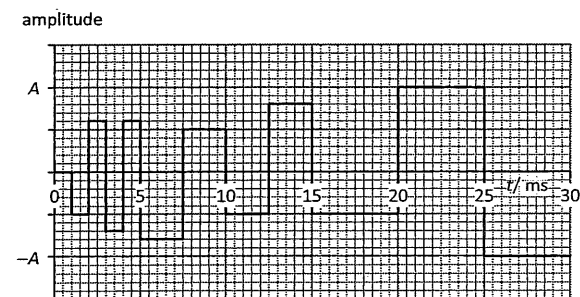


Fig. 7.1

- On Fig. 7.2, sketch the anti-noise waveform for $0 \text{ ms} \leq t \leq 30 \text{ ms}$. [2]

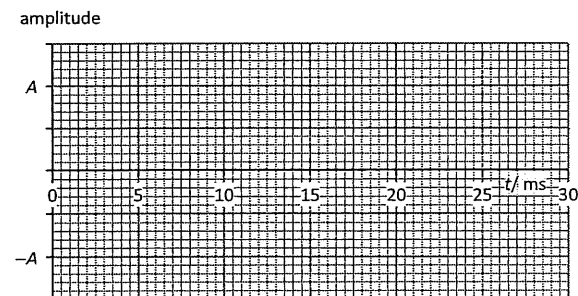


Fig. 7.2

- ii. The final resultant waveform reaching the ears is shown in Fig. 7.3.

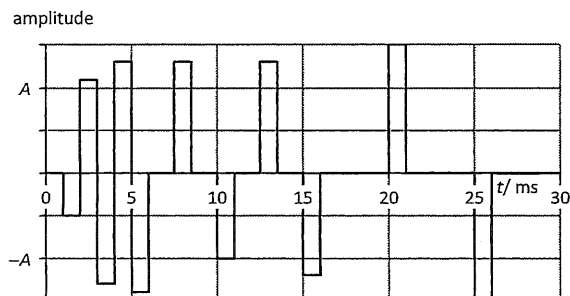


Fig. 7.3

Using Fig. 7.3, suggest if Active Noise Control is better suited for noises of higher or lower frequencies. [2]

[Total: 6]

8. [EJC/H2/Prelims 2018/P3/7* Modified]

- a. i. Explain what is meant by an *alternating current*. [1]
 ii. Fig. 8.1. shows a metal wire held taut between two knife edges X and Y. The wire passes between opposing poles of permanent bar magnets.

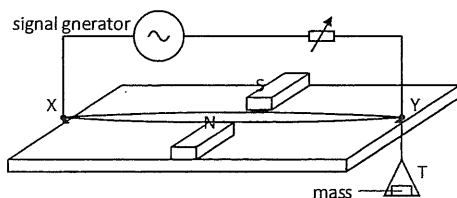


Fig. 8.1

The wire vibrates when a sinusoidal alternating source is connected across the wire. Explain how these vibrations are created. Describe the properties of these vibrations. [4]

- iii. When the frequency of the alternating source is 120 Hz, a standing wave in its fundamental mode is observed in the wire. XY is 2.0 m in length.
 1. Calculate the speed of the wave in the wire. [2]
 2. Explain, with reference to the formation of a stationary wave, what is meant by the speed calculated in a.iii.1. [4]
 iv. When the mass hanging below pulley P is doubled, the fundamental mode standing wave is observed to occur when the frequency of the alternating source is 170 Hz. When the mass was triple that of the original set up, the fundamental mode frequency is 208 Hz.
 Suggest if the relationship

$$v \propto \sqrt{T}$$

is valid, where v is the wave velocity in the metal wire, and T is the tension in the metal wire. [3]

- b. i. State what is meant by the *diffraction* of a wave. [1]

- ii. Light of wavelength 633 nm from a laser is directed normally at a diffraction grating, as illustrated in Fig. 8.2.

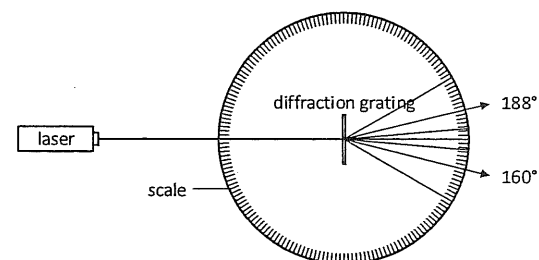


Fig. 8.2

The diffraction grating is situated at the centre of a circular scale, marked in degrees. The readings on the scale for the second order diffracted beams are 160° and 188°.

Calculate the number of lines per unit length of the slits in the diffraction grating. [4]

- iii. Suggest why the non-central fringes produced by light passing through a diffraction grating is brighter than that from the same source with a double slit. [1]

[Total: 20]

9. [HCI/H2/Prelims 2018/P3/3 Modified]

- a. State the principle of superposition. [2]
 b. A source of microwaves is placed on a table at a fixed distance from a detector. A vertical reflecting plate is placed a distance y from the source and the detector. Reflection at the reflector causes a phase change of π rad to the microwave.

Fig. 9.1 shows the view from above.

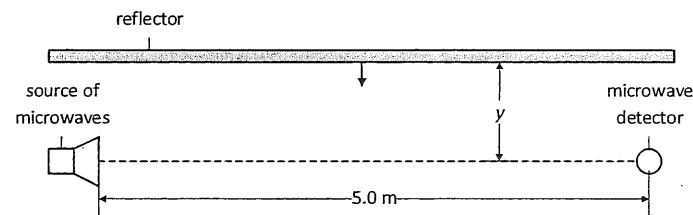


Fig. 9.1

The reflector is moved gradually towards the source and the detector, in the direction indicated by the arrow on the reflector in Fig. 9.1. The intensity measured by the detector alternates between high and low.

Explain these observations. [3]

- c. The reflector is fixed at a position $y=2.5$ m and the detector is at a distance 5.0 m away. The wavelength of the microwave is 0.59 m.
 i. Explain whether the intensity at the detector is a maximum or a minimum. [3]
 ii. Describe how the intensity of reception varies as the detector is moved towards the source along the dotted line until it reaches the source. [3]

[Total: 11]

10. [HCI/H2/Prelims 2018/P3/4]

This question is about an experiment to estimate the resolution of the human eye.

Two vertical parallel lines are drawn on a piece of card. The separation between the lines y is 2.0 ± 0.5 mm. The card is fixed to a wall at eye level.

A group of students look at the card whilst each covering one eye. They walk back from the card until they can no longer separate the two lines. The distance L between the eye and the card is measured.

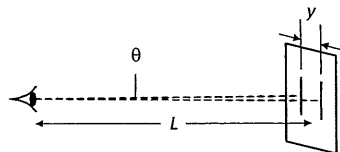


Fig. 10.1

The results from the five students are shown in Fig. 10.2.

student	A	B	C	D	E
maximum distance L / m	6.2	5.8	6.1	5.9	6.1

Fig. 10.2

- a. i. State the uncertainty of the distance L based on students' results. [1]
- ii. A student suggests that the uncertainty in the distance L can be ignored when calculating the minimum angle θ that can be resolved because of the uncertainty in the separation of the lines y on the card. Comment on this suggestion, explaining whether or not you agree. [3]
- iii. Calculate a value for the minimum angle (with its corresponding uncertainty) that can be resolved. [3]
- b. The diameter of the pupil of the eye is estimated to be 3 mm. The two lines are marked using red ink.
 - i. Explain what is meant by the Rayleigh criterion for the resolution of two patterns. [2]
 - ii. Estimate the minimum angle using the Rayleigh criterion. [2]

[Total: 11]

11. [JC/H2/Prelims 2018/P3/5 Modified]

Fig. 11.1 shows a demonstration of a *stationary wave* on a string.

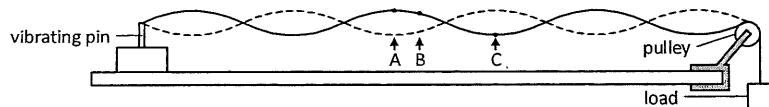


Fig. 11.1

- a. i. State whether the following pairs of points on the string are vibrating *in phase*, *in antiphase*, or neither in phase nor in antiphase.
 1. A and B
 2. A and C
- ii. Describe briefly how you could check this experimentally.

[2]

[1]

- b. i. Progressive waves transfer energy through the medium; stationary waves do not do this. Describe the difference in term of the amplitude of the wave between progressive and stationary waves. [2]
- ii. Explain how, in the set-up as shown in Fig. 11.1, the stationary wave can be thought of as arising from progressive waves. [2]

[Total: 7]

12. [JJC/H2/Prelims 2018/P2/5 Part]

- b. In Fig. 12.2, a narrow beam of monochromatic light of wavelength 590 nm passes through two different diffraction gratings A and B positioned perpendicular to each other. Assume that the distance between the two gratings is negligible.

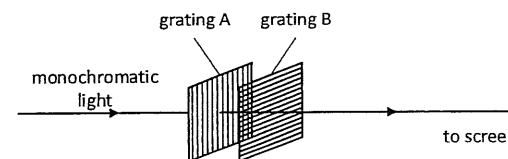


Fig. 12.2

As a result of this arrangement, a diffraction pattern is formed on a screen placed 2.50 m away from the gratings. The pattern is shown in Fig. 12.3.

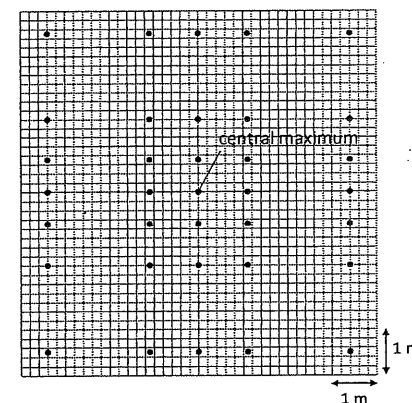


Fig. 12.3

- i. Explain the meaning of the term diffraction. [1]
- ii. By taking appropriate measurements from Fig. 12.3, determine the number of lines per unit length on diffraction grating B. [3]
- iii. Based on the diffraction pattern shown in Fig. 12.3, explain qualitatively what can be deduced about the slit separation of both diffraction gratings. [2]
- iv. The two diffraction gratings are now replaced with a fine nylon mesh with 24000 nylon threads per metre of the mesh. Suggest one way in which the diffraction pattern observed would be different. [1]

[Total: 7]

13. [MI/H2/Prelims 2018/P2/4]

- a. A contractor tries to measure the depth of a new well shaft to build a ladder to reach the bottom of the shaft. He uses an audio oscillator with adjustable frequency and applies it across the top of the well. Two successive resonances are heard at 70.6 Hz and 90.8 Hz. The speed of sound is 343 m s^{-1} . Determine the depth of the well. [3]
- b. An engineering student designed an Automated Guided Vehicle (AGV) using interference of radio waves from two coherent emitters 2.0 m apart emitting radio waves of frequency f that are in phase as shown in Fig. 13.1. The computer on the AGV detects and searches for lines of constructive interference and adjusts the AGV so that it is always aligned along the centre line, in the middle of the emitters.

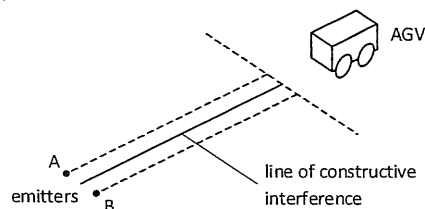


Fig. 13.1

- i. State and explain one disadvantage in using this method for the AGV to align itself along the centre-line. [2]
- ii. During one such operation, the AGV strays off the centre-line as shown in Fig. 13.2.

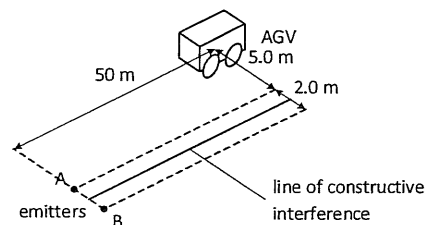


Fig. 13.2

Fig. 13.3 shows the radio signals X and Y detected by the receiver on the AGV.

signal captured by AGV

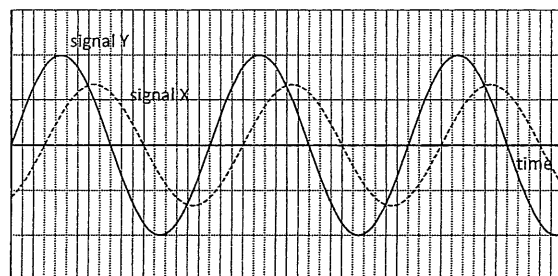


Fig. 13.3

- State and explain whether the source of signal X is from emitter A or B. [2]
- State the phase difference between signals X and Y. [1]
- Assuming the path difference is less than one wavelength, determine frequency f of the radio wave used. [4]

[Total: 12]

14. [MJC/H2/Prelims 2018/P2/3 Part]

- a. State two differences between stationary waves and progressive waves. [2]
- b. Same as [DHS/H2/Prelims 2018/P2/4]
- c. A cord is held under tension between two fixed points A and B, as shown in Fig. 14.3. The distance AB is 0.400 m.

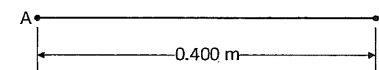


Fig. 14.3

- Explain why only stationary waves of certain frequencies are able to form between A and B. [1]
- The string is made to resonate in a mode with the third lowest possible frequency. Calculate the wavelength of this wave. [1]
- By reference to the formation of the stationary wave, explain the significance of the product of frequency and wavelength for a stationary wave. [2]

[Total: 6]

15. [NJC/H2/Prelims 2018/P3/7 Modified]

- State and explain the effect of the width of aperture on the diffraction of the wave. [2]
 - The band is practising in the band room down the corridor with the door left slightly ajar. Explain why notes of a certain range of frequencies can be heard more clearly than others at the other end of the corridor. [2]
- A plane diffraction grating is illuminated normally by a monochromatic light source of wavelength λ . Light rays from each slit travel a different distance before arriving at a particular point on the screen. In order to derive an expression for the path difference, the light rays that meet at the same point on the screen can be approximated as parallel rays.

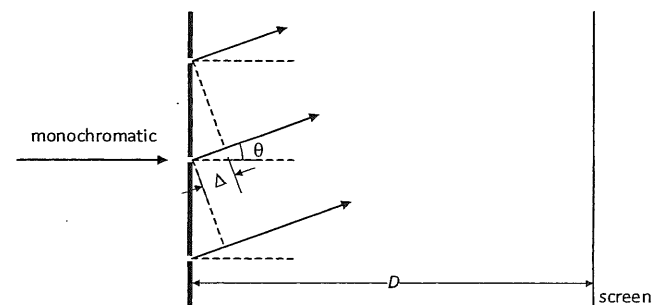


Fig. 15.1

- State the condition where the rays can be assumed to be parallel. [1]
 - Derive an expression for the path difference Δ between adjacent rays in terms of p and θ , where p is the number of lines per unit length of the grating, and θ is the angle between the emergent ray and the normal. [2]
 - Hence, derive an expression relating p and θ for the constructive interference of the rays. Define any additional symbol you use. [2]
- c. A diffraction grating is set up at the centre of a rotating table which completes a revolution every 3.0 s. The grating is illuminated normally by monochromatic light of wavelength λ from a source which is also mounted on the table as shown in Fig. 15.2.

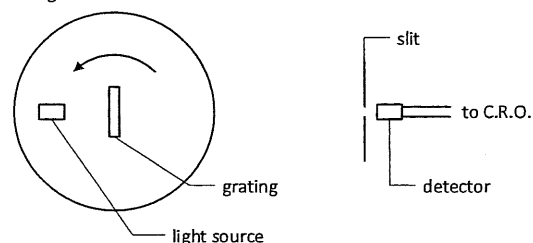


Fig. 15.2

The emergent beams of light from the grating are monitored by means of a stationary opto-electrical detector. The output from detector is displayed on a cathode ray oscilloscope (c.r.o.). With the time-base set at 0.10 s cm^{-1} , the trace obtained is shown in Fig. 15.3. The relative positions of the peaks are as indicated.

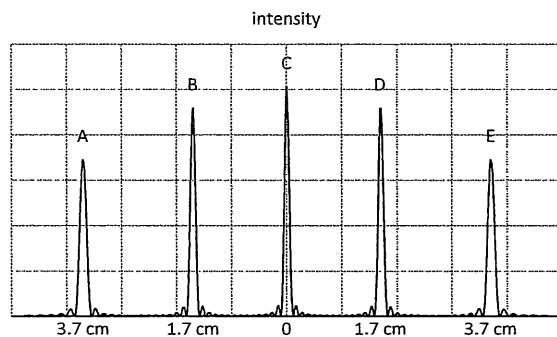


Fig. 15.3

- Calculate the angular speed of rotation of the grating. [1]
- Explain the appearance of the trace in Fig. 15.3. [2]
- Calculate the angle θ in radians, as defined in b.ii., for peaks D and E. [3]
- Using peak E, calculate the wavelength of the light if the grating has 550 lines per mm. [2]
- Explain why it is preferable to calculate the wavelength using peak E rather than peak D. [1]
- Sketch in Fig. 15.3, the c.r.o. display, if the diffraction grating is replaced by a double slit of the same slit separation as the diffraction grating. [2]

[Total: 20]

16. [NYJC/H2/Prelims 2018/P2/4]

- Explain what is meant by polarised light. [1]
- Explain why two coherent sources of light that are polarised in planes perpendicular to each other will not produce interference fringes. [1]
- Fig. 16.1 shows the setup for a Young's double slit experiment. The polarisers are initially aligned such that the light from the slits is polarised in parallel planes. The slit separation of the double slit is 0.10 mm, the distance between the double slit and the screen is 1.5 m and the separation of the fringes produced is 0.90 cm.

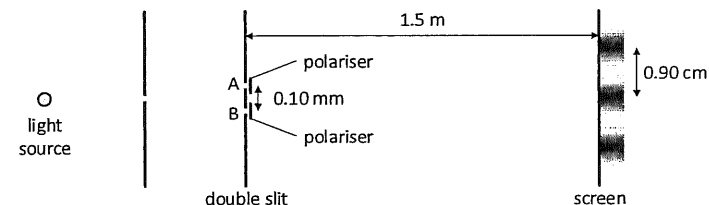


Fig. 16.1

- Calculate the wavelength of the light source. [2]
- One of the polarisers is rotated by an angle of 45° about an axis parallel to the incident light. Describe the appearance of the fringes with reference to its original appearance. [2]

[Total: 6]

17. [NYJC/H2/Prelims 2018/P3/8 Modified]

- Point source P, consisting of light with wavelength 630 nm, passes through a narrow slit and is incident on a screen at a distance of 2.4 m from the slit. Fig. 17.1 below shows the variation with distance x along the screen of intensity I of the light.

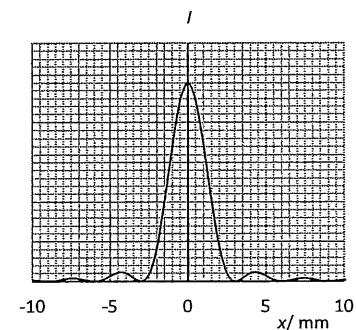


Fig. 17.1

- Use Fig. 17.1 to determine the width of the slit. [2]
 - State the effect on the pattern on the screen if each of the following changes is made separately. [2]
- The width of the single slit is reduced,
 - The red source is replaced with another source of violet light.

[2]

[2]

- b. Another point source Q, emitting light of the same wavelength as P, is placed 5.0 mm from P, and the two sources are arranged as shown in Fig. 17.2. The slit is now replaced with another slit of width b .

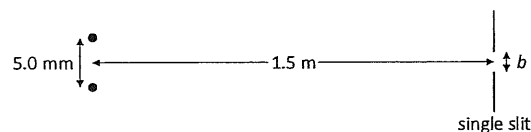


Fig. 17.2

- State the condition for the image of P and the image of Q formed on the screen to be just resolved. [1]
 - Sketch, on Fig. 17.1, the graph for point source Q when the condition in b.i. is met. [2]
 - Calculate the minimum width b of the slit for the two images to be just resolved. [3]
- c. Two microwave transmitters, A and B, are separated by a distance of 0.80 m. They each transmit a microwave of frequency 3.00 GHz.

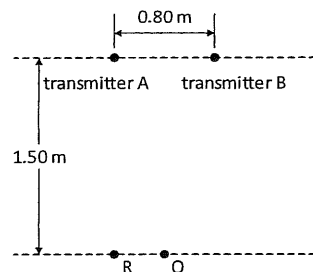


Fig. 17.3

A microwave receiver is moved along a line parallel to, and 1.50 m from, the line joining the two transmitters, as shown in Fig. 17.3. Point O lies on the line where it is equidistant from both transmitters, while point R is at a distance 1.50 m from transmitter A. As the receiver moves from O to R, a minimum signal is detected at R.

- Determine the path difference between the two waves arriving at R. [2]
- Show that the two waves are emitted in antiphase at the transmitters. [3]

Fig. 17.4 below shows the displacement-time graph of the waves arriving at R when only transmitter A is switched on and when both transmitters A and B are switched on.

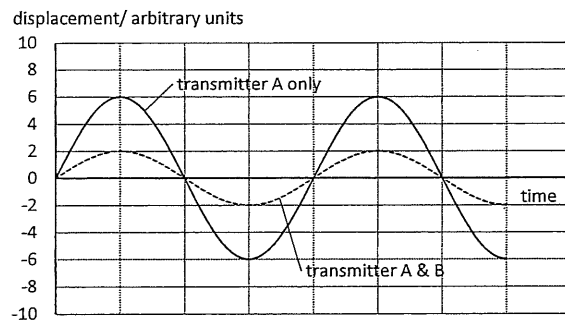


Fig. 17.4

- Sketch, on Fig. 17.4, the displacement-time graph of the waves arriving at R when only transmitter B is switched on. [1]

- Determine the ratio, at point R, of $\frac{\text{intensity of microwave due to transmitter A}}{\text{intensity of microwave due to transmitter B}}$. [2]

[Total: 20]

18. [PJC/H2/Prelims 2018/P3/4 Modified]

- State the principle of superposition. [2]
- Fig. 18.1 shows the variation with time t of the displacements x_A and x_B of the particles at a point P due to sound waves A and B.

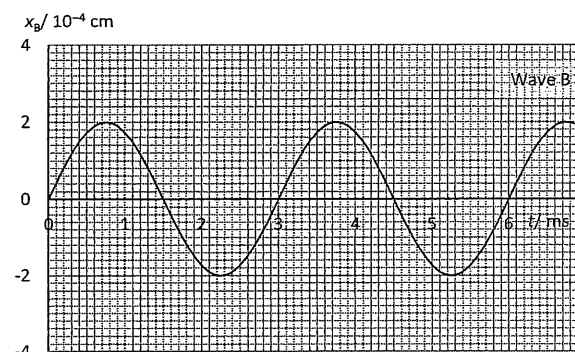
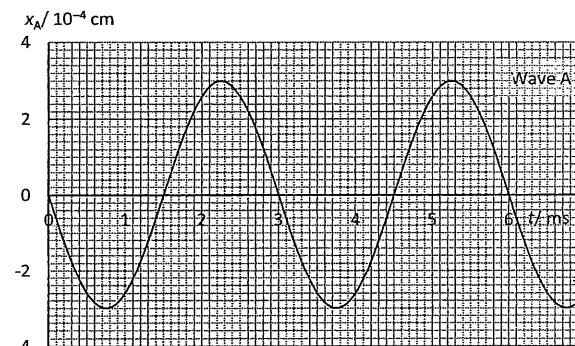


Fig. 18.1

- Determine the resultant displacement for the two waves at point P at time $t = 4.0$ ms, [1]
- The intensity of wave A alone at point P is I . Calculate the resultant intensity, in terms of I , due to the two waves at point P. [2]

- c. Fig. 18.2 shows the two loudspeakers that produced the two sound waves A and B as mentioned in (b). Point O is equal distance from both loudspeakers.



Fig. 18.2

- A detector is moved from point O towards Y along the line XY. After detecting three maxima including point O, a third minima is located at point P, which is 30.2 m from point O. Determine the speed of sound in air. [3]
- Fig. 18.3 shows the two loudspeakers rotated to face each other but still kept 3.0 m apart. The sound emitted by the loudspeakers are in anti-phase.

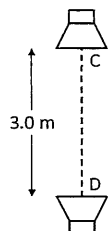


Fig. 18.3

If the detector is moved along the line CD, describe the variation of the sound intensity detected. [1]

[Total: 9]

19. [RVHS/H2/Prelims 2018/P3/7 Modified]

- a. A glass tube, open at both ends, has fine dust sprinkled along its length. A speaker is placed near one open end of the tube, as shown in Fig. 19.1.

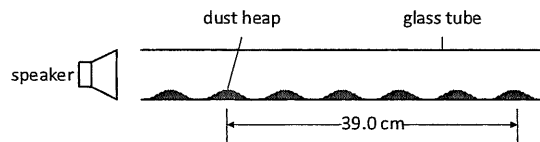


Fig. 19.1

- On Fig. 19.1, sketch the form of the stationary wave set up in the tube and mark, with the letter A, the position(s) of any displacement antinode(s) of the stationary wave. [2]
- The distance between six heaps is 39.0 cm. Show that the wavelength of the stationary wave is 15.6 cm. [1]
- The frequency at which these heaps are formed is 2.14 kHz. Calculate the speed of sound in the tube. [2]
- Calculate the lowest frequency of the sound that produces stationary wave in the tube. [3]
- State and explain the change, if any, to the dust heaps if the right open end of the tube is blocked with a lid, keeping the frequency at 2.14 kHz. [2]

- b. Fig. 19.2 shows an arrangement used to determine the speed of microwaves in the laboratory.

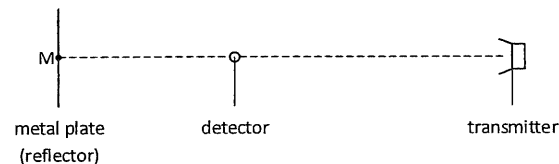


Fig. 19.2

A stationary wave is formed between the metal plate and the transmitter. As the detector is moved along a line between the transmitter and the point M, it registers a series of high intensity signals. The positions of these high intensity signals are marked by crosses on an axis as shown in Fig. 19.3.

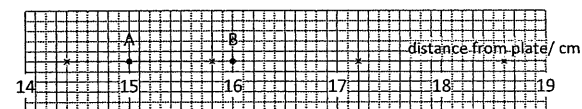


Fig. 19.3

- State the phase difference, in radians, between points A and B. [1]
- The transmitter emits microwaves of frequency 10 GHz. Use Fig. 19.3 to deduce an experimental approximation for the speed of the microwaves. [2]
- The detector is now fixed at a position of high intensity. If the metal plate is moved away from the transmitter at a speed of 1.00 cm s^{-1} while the detector is kept stationary, determine the frequency of fluctuation of the signal registered on the detector. [2]

The metal plate is now removed from the arrangement and a sheet of hardboard is inserted between the detector and the transmitter without either of these being moved as shown in Fig. 19.4. The hardboard is a soft boundary that partially reflects microwaves.

When the hardboard is inserted, the intensity of the wave drops to $1/4$ th of its original value.

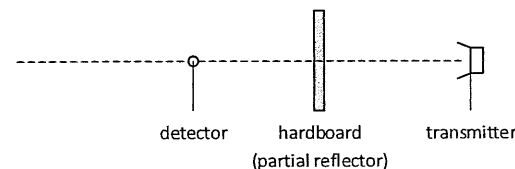


Fig. 19.4

- iv. Calculate the ratio of $\frac{\text{amplitude of detected wave after hardboard insertion}}{\text{amplitude of detected wave before hardboard insertion}}$. [2]

The metal plate is now placed back in original position as shown in Fig. 19.5.

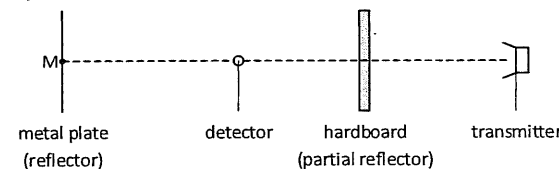


Fig. 19.5

- v. Fig. 19.6 shows the variation of intensity with position as the detector is moved between the transmitter and M at a particular frequency before the hardboard is inserted.

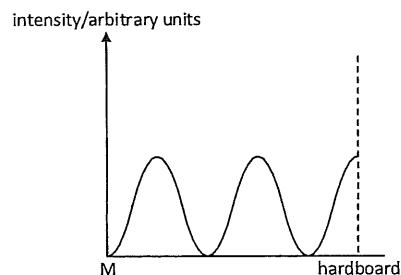


Fig. 19.6

Sketch on Fig. 19.6 the variation of intensity with position as the detector is moved between the hardboard and M after the hardboard is inserted.

[3]

[Total: 20]

20. [SRJC/H2/Prelims 2018/P2/4 Modified]

- a. Fig. 20.1 shows a string stretched between two fixed points P and Q.

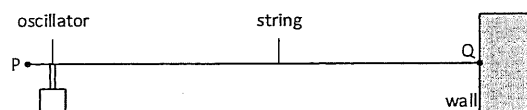


Fig. 20.1

An oscillator is attached to the string near P. The oscillator has a frequency of 50.0 Hz.

The stationary wave produced on PQ at an instant time t is shown in Fig. 20.2. Each point on the string is at its maximum displacement.

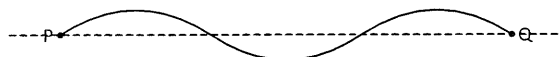


Fig. 20.2

- On Fig. 20.2, label all the nodes with the letter N and the antinodes with the letter A along the dotted line PQ.
 - On Fig. 20.2, draw the stationary wave at $t + 5.0$ ms.
- b. Sound waves is directed from a loudspeaker towards a metal plate. A microphone, connected to a cathode ray oscilloscope (CRO), is placed in between the loudspeaker and the metal plate as shown in Fig. 20.3.

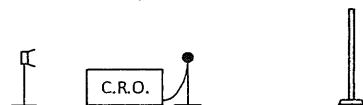


Fig. 20.3

- Explain how stationary waves are formed in between the loudspeaker and the metal phase.
- The trace on the CRO in Fig. 20.4 shows the variation of signal with time at an antinode. The time-base setting on the CRO is 0.10 ms cm^{-1} .

[3]

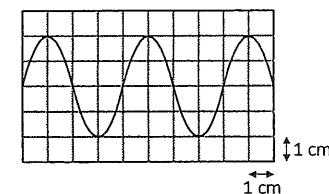


Fig. 20.4

The microphone is then moved by 3.3 cm and the trace on the CRO now records zero amplitude.

- Determine the frequency of the sound.
- Calculate the speed of sound.

[1]

[2]

[Total: 8]

21. [SRJC/H2/Prelims 2018/P3/3]

- a. As seen in Fig 21.1, a monochromatic light of wavelength 580 nm is used to produce an interference pattern on screen AB. The separation between the slits is 0.41 mm and the distance between the double slit and the screen is D . Point Y is at a distance $x = 2.0$ mm from point O and it is the position of the first dark fringe. The intensities of the light passing through the two slits are the same.

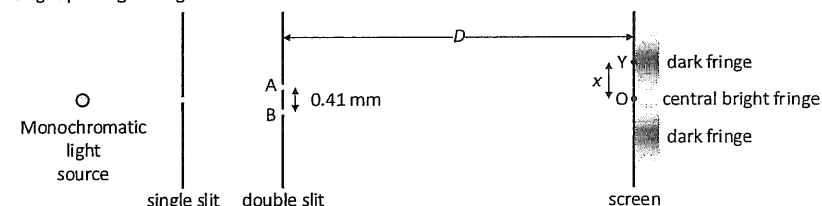


Fig. 21.1

- Calculate the path difference between the 2 waves arriving at point Y from the slits.
- Calculate the distance D .
- The width of both slits is reduced by the same amount without altering their separation. The original variation with distance x from point O of the intensity is as shown in Fig. 21.2.

[1]

[2]

intensity

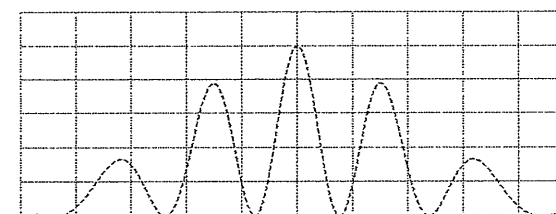


Fig. 21.2

Sketch, on Fig. 21.2, the new variation of intensity.

[3]

- b. A diffraction grating is used to measure the wavelengths of light. The angle θ of the second order maximum is measured for each wavelength. The variation with wavelength λ of $\sin \theta$ is shown in Fig 21.3.

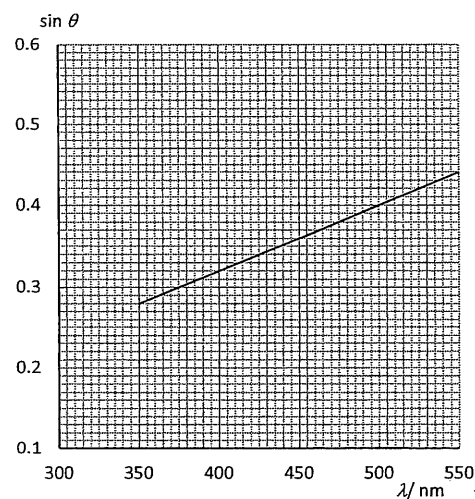


Fig. 21.3

- Calculate the slit separation d of the diffraction grating. [3]
- On Fig. 21.3, sketch a line to show the results that would be obtained for the first order maxima. [1]

[Total: 10]

22. [TPJC/H2/Prelims 2018/P3/7 Modified]

- Explain what is meant by the diffraction of a wave. [1]
- A ripple tank is used to show the diffraction and interference of waves. On Fig. 22.1, plane wavefronts are shown in two cases, approaching a single slit.

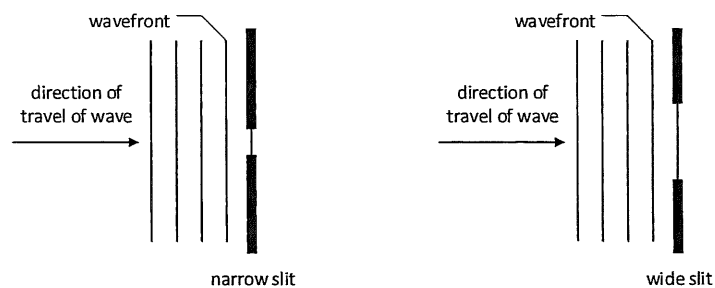


Fig. 22.1

Draw four wavefronts to show the waves after they have passed through the slit in each case.

[3]

- In a separate experiment using double-slit arrangement, two source interference of water waves were not observed when the slit widths are wide.

However, narrowing the slits while maintaining the same slit separation produced observable interference patterns.

Suggest why. [3]

- By reference to waves, state what is meant by coherent sources. [2]
 - Two sources S_1 and S_2 operate in phase to produce circular waves as shown in Fig. 22.2. The circles represent the crests of the waves.

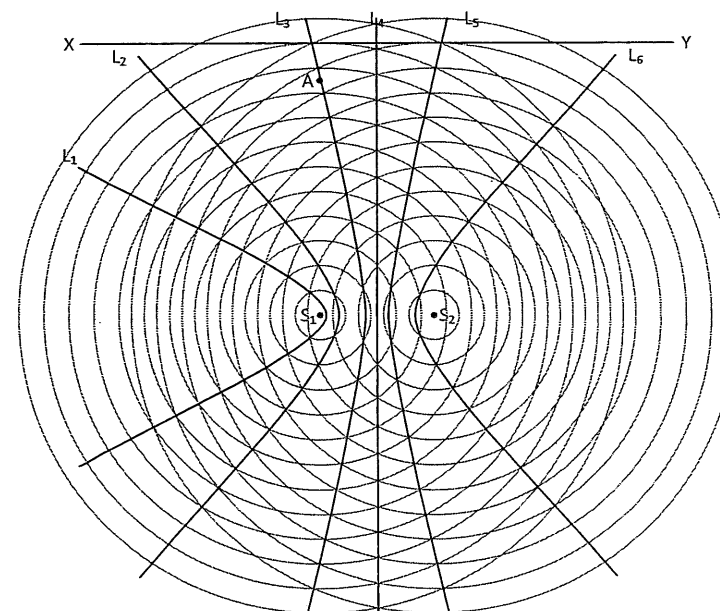


Fig. 22.2

Along the line XY, there is a series of alternate maxima and minima.

- Explain what lines L_1 to L_6 represent and whether point A, as indicated in Fig. 22.2, is a point of constructive or destructive interference. [2]
- State the relationship between the magnitude of the path difference $S_1A - S_2A$ and the wavelength. [1]
- Draw, on Fig. 22.2, a line to show one direction along which the waves have minimum amplitude. Label this line as Z. [1]
- If the separation of the sources is increased, state what would happen to the spacing between maxima. [1]

- c. Monochromatic light is passed through a rectangular slit of width b of 0.20 mm. The light is observed on a screen placed 0.75 m from the slit, as shown in Fig. 22.3.

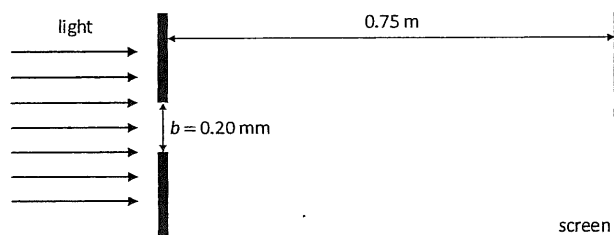


Fig. 22.3

Light passing through the slit is diffracted through an angle θ .

The variation with angular position θ of the intensity I of the light is shown in Fig. 22.4. The maximum value of intensity is I_0 .

The angular separations between successive minima are the same.

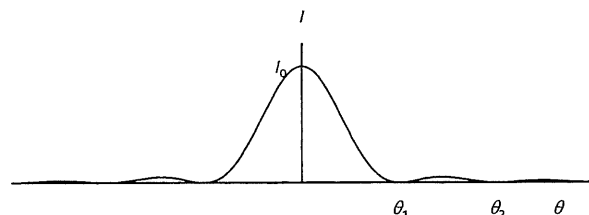


Fig. 22.4

- The value of θ_1 , the angle of diffraction corresponding to the first order minima, is 0.17° . Determine the wavelength, in nm, of the monochromatic light. [2]
- Calculate the width of the central fringe, as observed on the screen. [2]
- Sketch, in Fig. 22.4, how the variation with angular position θ of the intensity I if the slit width is halved. [2]

[Total: 20]

23. [VIC/H2/Prelims 2018/P3/3 Modified]

Two identical speakers placed at positions A and B are connected to the same signal generator, 1.2 m apart. Meena walks in a direction parallel to AB, 20 m away, as shown in Fig. 23.1.

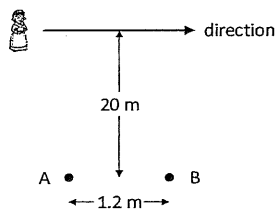


Fig. 23.1 (not to scale)

As she walks, she detects a series of loud and soft sounds.

- Adjacent loud regions are 1.5 m apart. Calculate the frequency of the sound used given that the speed of sound is 340 m s^{-1} . [3]
- At one of the loud regions, Meena detects an intensity of I . The amplitude of wave from each speaker is A . State the resultant wave amplitude at Meena's location. [1]
- The power output of one of the speakers is reduced to half its original value, while the other speaker remains unchanged.
 - For the speaker with the reduced power output, state the amplitude of its wave at Meena's location in terms of A . [1]
 - Calculate the new resultant wave amplitude at Meena's location in terms of A . [1]
 - Calculate the new intensity detected by Meena in terms of I . [2]

[Total: 8]

24. [YJC/H2/Prelims 2018/P3/8 Modified]

- A student performed an experiment to determine the speed of sound. She stands at point P, a distance d from the brick wall, and claps her hands. A microphone on her is connected to a timing device and records the time t between the clap and its echo. The experiment is performed at three distances d and the results are plotted in Fig. 24.1.

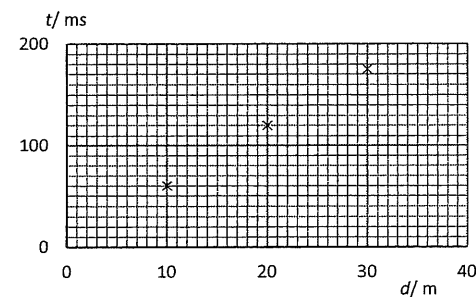


Fig. 24.1

Determine a value for the speed of sound. Explain your working. [3]

- In a second experiment, as shown in Fig. 24.2, the student sets up two small loudspeakers, S_1 and S_2 , connected to the same signal generator, set to 8300 Hz.

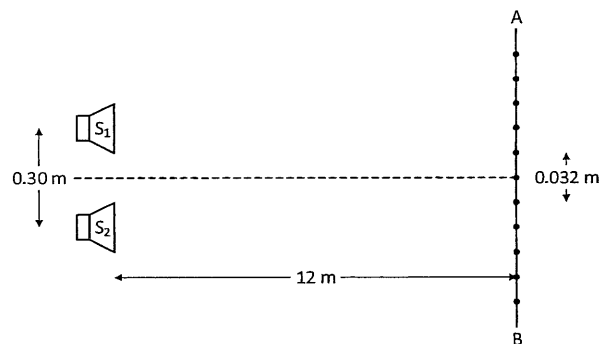


Fig. 24.2 (not to scale)

She moves a microphone along the line AB and locates alternating maxima and minima at the positions along AB. The maxima are located at the dots.

- i. Determine the wavelength of the sound produced from S_1 and S_2 . [2]
 - ii. Hence, calculate the speed of sound from this experiment. [1]
 - iii. 1. Label with a letter 'M' on one of the dots where the path difference is $S_2M - S_1M = 2\lambda$. [1]
 2. Explain why 'M' is a maxima. [1]
 - iv. When the signal generator is set to 300 Hz, the student does not find a succession of maxima and minima as the microphone is moved along the line AB. Explain why this happens. [3]
- c. Fig. 24.3 shows an arrangement used to analyse the light emitted by a source. The light source emits a range of wavelengths from 500 nm to 700 nm. The light is incident on a diffraction grating that has 10000 lines per metre. The diffracted pattern is formed on the screen placed at a distance away from the grating. OX is the line that indicates the direction of the undiffracted light.

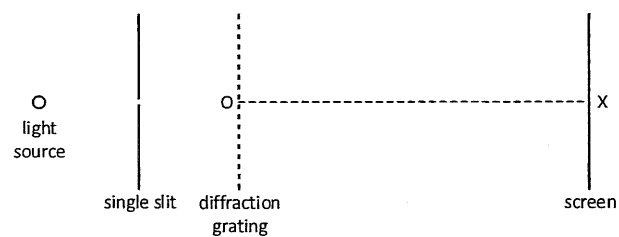


Fig. 24.3

- i. Calculate the angle from OX at which the first order maximum for the light of wavelength of 500 nm is formed. [2]
- ii. Calculate the angle subtended by the first order spectrum on the same side of OX. [2]
- iii. Calculate the width of the first order spectrum if the screen is placed at 2.0 m from the diffraction grating. [2]
- iv. The single slit is initially illuminated by light from a point source that is 0.020 m from the slit.
 State and explain how the intensity of light incident on the single slit changes when the light source is moved to a position 0.050 m from the slit. [3]

[Total: 20]

